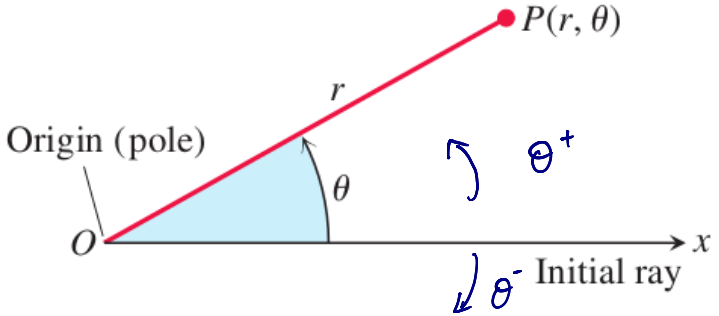
## 10.3 Polar Coordinates

### Definition of Polar Coordinates




Fix an **origin** $O$ **(pole)** and an **initial ray** from $O$. Then every point $P$ on the plane can be located by assigning to it a **polar coordinate** $(r, \theta)$ with

$r$ = the directed distance from $O$ to $P$,
$\theta$ = the angle from initial ray to ray $OP$.

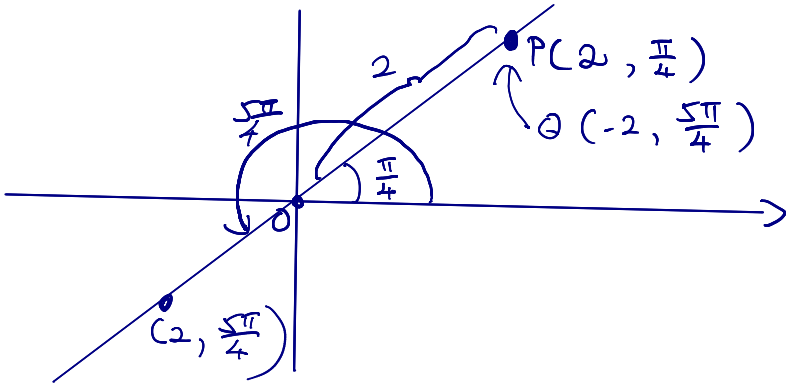
**Polar Coordinates** $P = P(r, \theta)$

Here, we note that $\theta$ is positive when measured counterclockwise, and negative when measured clockwise.

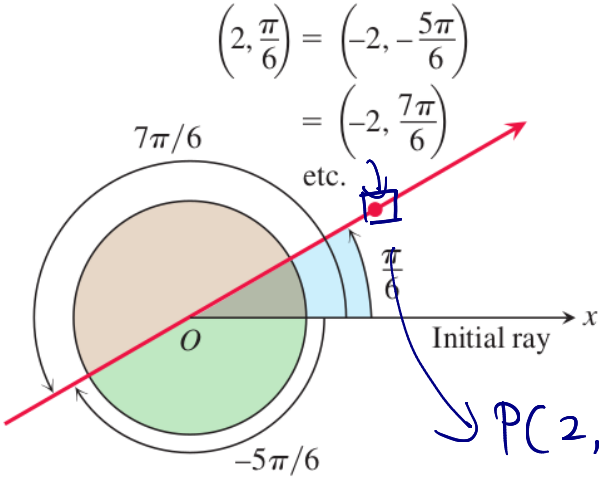
Sometimes, we also allow $r$ to be negative as well - when we want to indicate that the ray is *going backward*.

$r > 0$ P is in the same quadrant as $\theta$.

$r < 0$ P is in the opposite side of the pole.



Example: Polar coordinates of the point $P(2, \pi/6)$.



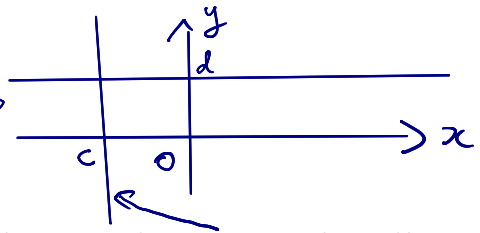

$P(2, \frac{\pi}{6}) = P(2, -\frac{11\pi}{6}) = P(-2, \frac{7\pi}{6})$

$P(2, \frac{\pi}{6}) = P(2, \frac{\pi}{6} + 2n\pi)$ for any $n \in \mathbb{Z}$

$= P(-2, \frac{7\pi}{6} + 2n\pi)$ for any $n \in \mathbb{Z}$.

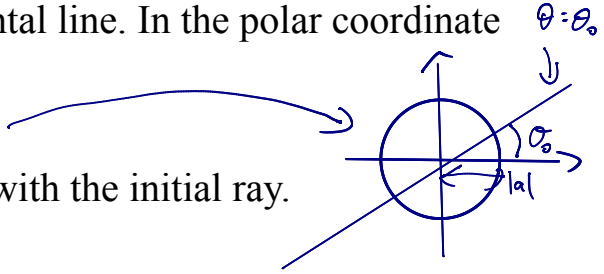
**Polar Equations and Graphs**

A graph can be specified by a polar equation




Example: Recall that in the Cartesian coordinates system, the equation $x = c$ describes a vertical line and the equation $y = d$ describes a horizontal line. In the polar coordinate system, the equations

1. $r = a$ is the circle of radius $|a|$ centred at $O$
2. $\theta = \theta_0$ is the line through $O$ making an angle $\theta_0$ with the initial ray.



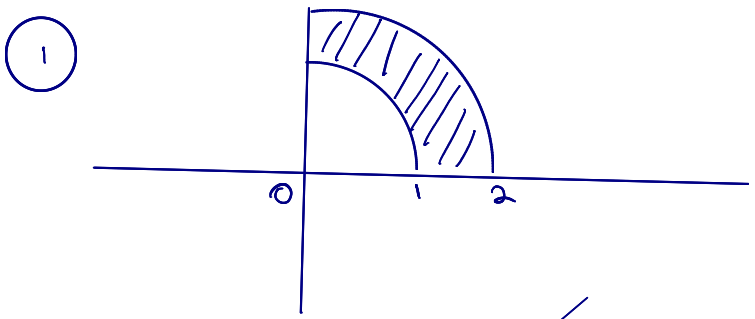

Examples:

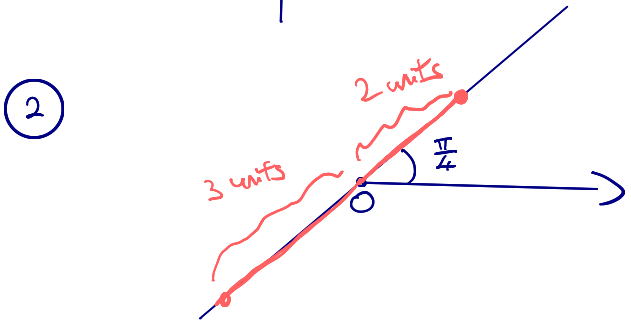
1. $r = 1$ and $r = -1$ are the equations representing the same graph, that is, the circle of radius 1 centred at $O$.
2. $\theta = \pi/6$, $\theta = 7\pi/6$, and $\theta = -5\pi/6$ all represent the same line, the line in the previous example.

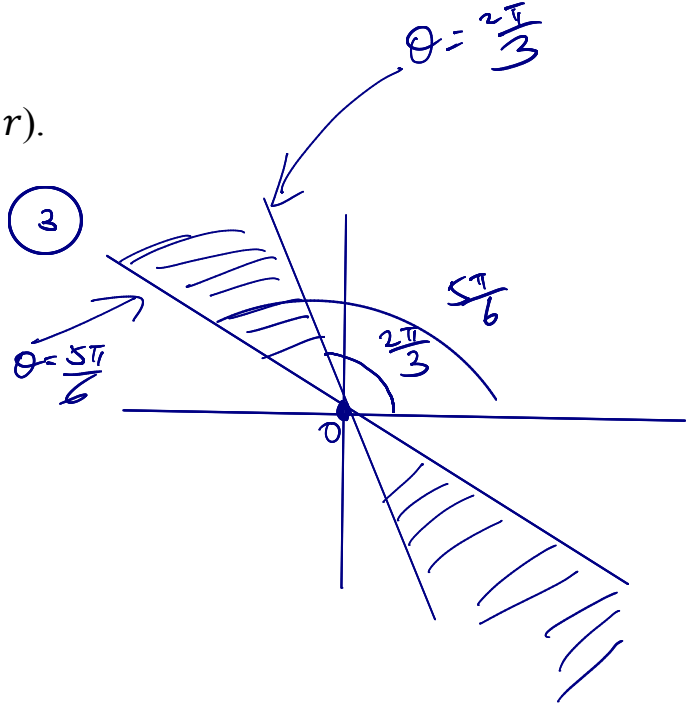
We can also define regions using polar equations.

Examples:

1. $1 \leq r \leq 2$ and $0 \leq \theta \leq \frac{\pi}{2}$.
2. $-3 \leq r \leq 2$ and $\theta = \frac{\pi}{4}$.
3. $\frac{2\pi}{3} \leq \theta \leq \frac{5\pi}{6}$ (no restriction on $r$).



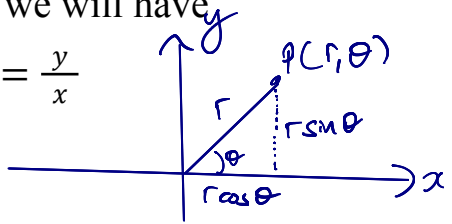



**Relating Polar and Cartesian Coordinates**

In both systems, we place both the origins together and take the initial polar ray as the positive $x$-axis. The $y$-axis is then the ray $\theta = \pi/2,\ r > 0$. Then we will have

$$x = r\cos\theta \qquad y = r\sin\theta \qquad r^2 = x^2 + y^2 \qquad \tan\theta = \frac{y}{x}$$

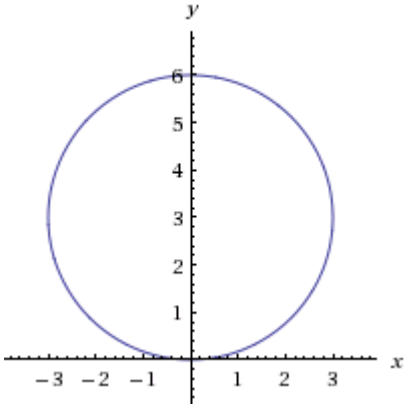



$P(r,\theta) = (r\cos\theta, r\sin\theta)$

Example:

| Polar equation | Cartesian equivalent |
|---|---|
| $r\cos\theta = 2$ | $x = 2$ |
| $r^2\cos\theta\sin\theta = 4$ | $xy = 4$ |
| $r^2\cos^2\theta - r^2\sin^2\theta = 1$ | $x^2 - y^2 = 1$ |
| $r = 1 + 2r\cos\theta$ | $y^2 - 3x^2 - 4x - 1 = 0$ |
| $r = 1 - \cos\theta$ | $x^4 + y^4 + 2x^2y^2 + 2x^3 + 2xy^2 - y^2 = 0$ |

Example: Find a polar equation for the circle $x^2 + (y-3)^2 = 9$.




center = $(0,3)$

radius = $3$

$x^2 + (y-3)^2 = 9$

$(r\cos\theta)^2 + (r\sin\theta - 3)^2 = 9$

$\Rightarrow r^2\cos^2\theta + r^2\sin^2\theta - 6r\sin\theta = 0$

$\Rightarrow r^2 - 6r\sin\theta = 0$

$\Rightarrow r(r - 6\sin\theta) = 0$

$\Rightarrow r = 0$ OR $r = 6\sin\theta$.

Since $r \neq 0$ $\Rightarrow$ $\underline{\underline{r = 6\sin\theta}}$ ← the polar equation of the circle centered at $(0,3)$ with radius 3.

Examples: Find the cartesian equivalent of the following polar equations.

(a). $r\cos\theta = -4$ ⇒ $x = -4$, a vertical line with $x = -4$.

(b). $r^2 = 4r\cos\theta$

(c). $r = \frac{4}{2\cos\theta - \sin\theta}$

(b) $r^2 = 4r\cos\theta$

$\Rightarrow x^2 + y^2 = 4x$

$\Rightarrow x^2 - 4x + y^2 = 0$

$\Rightarrow (x-2)^2 + y^2 = 4$

→ a circle with center $= (2, 0)$ & radius 2.

(c) $r(2\cos\theta - \sin\theta) = 4$

$\Rightarrow 2r\cos\theta - r\sin\theta = 4$

$\Rightarrow 2x - y = 4$

$\Rightarrow y = 2x - 4$

→ a line with gradient 2 & y-intercept $= -4$